

Global Foliation Trace Route

This note records the sharp version of Plan 2: do not select one good near-boundary level. Use the whole outer family of level sets. The point is that the Holder/Serrin loss D_H is not just extra pointwise information; in level-set coordinates it is the radial kinetic energy of the foliation. Together with the isoperimetric loss D_I , it gives a trace estimate for the boundary asymmetry.

The argument below is written first in the coherent near-spherical foliation regime. This is the part that should replace the final selected-minimizer compactness step. The remaining task is an entry theorem saying that a small global level-set deficit puts the actual torsion levels into this weak foliation gauge on a fixed outer annulus.

Throughout this note normalize

$$|\Omega| = \omega_n, \quad B = B_1,$$

and write

$$\delta_T(\Omega) = E(\Omega) - E(B).$$

Let $u = u_\Omega$ be the torsion function,

$$E_t = \{u > t\}, \quad m(t) = |E_t|,$$

and introduce the volume radius

$$\rho = \left(\frac{m(t)}{\omega_n} \right)^{1/n}.$$

Equivalently, write $t = t(\rho)$ and $E_\rho = E_{t(\rho)}$. The exact level-set identity is

$$2\delta_T(\Omega) = \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt.$$

1 Coherent Outer Foliation Gauge

Fix $0 < \rho_* < 1$, for instance $\rho_* = 1/2$. Assume that for a.e. $\rho \in [\rho_*, 1]$ the regular level surface ∂E_ρ can be written in the modulated gauge

$$\partial E_\rho = \{z(\rho) + (\rho + h(\rho, \theta))\theta : \theta \in \partial B\},$$

with

$$\int_{\partial B} h(\rho, \theta) d\theta = 0, \quad \int_{\partial B} h(\rho, \theta) \theta_i d\theta = 0 \quad (i = 1, \dots, n).$$

The zero mode is removed by using the volume radius ρ . The first modes are removed by the moving center $z(\rho)$. This is important: translation is a neutral mode for Fraenkel asymmetry, so the center is allowed to move with ρ .

Define the non-neutral quadratic form

$$Q(h) := \int_{\partial B} (|\nabla_\theta h|^2 - (n-1)h^2) d\theta.$$

On the subspace orthogonal to constants and first spherical harmonics,

$$Q(h) \geq c_n \|h\|_{H^1(\partial B)}^2.$$

The coherent-foliation estimate needed from the geometry is

$$\boxed{\int_{\rho_*}^1 \left(\rho^{n+1} \|\partial_\rho h(\rho)\|_{L^2(\partial B)}^2 + \rho^{n-1} Q(h(\rho)) \right) d\rho \leq C_n \delta_T(\Omega).} \quad (\text{F})$$

Once (F) is known, sharp stability follows immediately.

2 Trace Estimate: No Boundary-Slab Loss

Since $\rho \in [\rho_*, 1]$, the weights ρ^{n+1} and ρ^{n-1} are bounded above and below by dimensional constants. For every non-neutral mode, the one-dimensional trace theorem gives

$$\|h(1)\|_{L^2(\partial B)}^2 \leq C_{\rho_*} \int_{\rho_*}^1 \left(\|\partial_\rho h(\rho)\|_{L^2(\partial B)}^2 + \|h(\rho)\|_{L^2(\partial B)}^2 \right) d\rho.$$

Using the coercivity of Q on the chosen gauge,

$$\boxed{\|h(1)\|_{L^2(\partial B)}^2 \leq C_n \int_{\rho_*}^1 \left(\rho^{n+1} \|\partial_\rho h(\rho)\|_2^2 + \rho^{n-1} Q(h(\rho)) \right) d\rho.} \quad (\text{T})$$

Combining (T) with (F) gives

$$\|h(1)\|_{L^2(\partial B)}^2 \leq C_n \delta_T(\Omega).$$

For a near-spherical set in the same volume and translation gauge,

$$\mathcal{A}(\Omega)^2 \leq C_n \|h(1)\|_{L^2(\partial B)}^2.$$

Hence

$$\boxed{\mathcal{A}(\Omega)^2 \leq C_n \delta_T(\Omega).}$$

This is the sharp Saint–Venant exponent. No parameter η is introduced: the trace is taken from a fixed annulus, and the radial derivative term supplied by D_H prevents the boundary deformation from being hidden in an arbitrarily thin layer.

3 Why D_I Gives the Angular Part

In the graph gauge,

$$P(E_\rho)^2 - P(B_\rho)^2 \geq c_n \rho^{2n-4} Q(h(\rho)) - \text{higher order}.$$

This is the Fuglede expansion of the perimeter deficit on the level ∂E_ρ , after removing the volume and translation modes.

Since $m(t(\rho)) = \omega_n \rho^n$, the denominator in the exact identity satisfies

$$n^2 \omega_n^{2/n} m(t(\rho))^{1-2/n} \simeq_n \rho^{n-2}.$$

In a near-spherical foliation,

$$-t_\rho(\rho) \simeq_n \rho.$$

Therefore

$$\int_{\rho_*}^1 \frac{D_I(t(\rho))}{n^2 \omega_n^{2/n} m(t(\rho))^{1-2/n}} (-t_\rho(\rho)) d\rho \geq c_n \int_{\rho_*}^1 \rho^{n-1} Q(h(\rho)) d\rho - \text{errors}.$$

For a sufficiently small graph radius, the errors are absorbed by the left-hand side or by the D_H contribution.

4 Why D_H Gives the Radial Kinetic Part

Let

$$X(\rho, \theta) = z(\rho) + (\rho + h(\rho, \theta))\theta$$

parametrize ∂E_ρ , and let ν_ρ be the outer unit normal to E_ρ . Define the normal velocity of the level foliation with respect to the volume radius:

$$V(\rho, \theta) := \partial_\rho X(\rho, \theta) \cdot \nu_\rho(\theta).$$

Since $u(X(\rho, \theta)) = t(\rho)$ and the outward normal is $\nu_\rho = -\nabla u/|\nabla u|$,

$$|\nabla u|(X(\rho, \theta)) = \frac{-t_\rho(\rho)}{V(\rho, \theta)}.$$

Thus the scalar factor $-t_\rho$ cancels out of the Holder deficit:

$$D_H(t(\rho)) = \left(\int_{\partial E_\rho} V d\mathcal{H}^{n-1} \right) \left(\int_{\partial E_\rho} \frac{1}{V} d\mathcal{H}^{n-1} \right) - P(E_\rho)^2.$$

Moreover, because ρ is the volume radius,

$$\int_{\partial E_\rho} V d\mathcal{H}^{n-1} = \frac{d}{d\rho} |E_\rho| = n\omega_n \rho^{n-1} = P(B_\rho).$$

So D_H is exactly the Cauchy deficit for the normal velocity V . In the graph gauge,

$$V = 1 + z'(\rho) \cdot \theta + \partial_\rho h + \text{quadratic terms}.$$

The moving center absorbs the first spherical harmonics, and the Cauchy deficit controls the variance of V . Projecting to the non-neutral modes gives

$$D_H(t(\rho)) \geq c_n \rho^{2n-2} \|\partial_\rho h(\rho)\|_{L^2(\partial B)}^2 - \text{lower order errors}.$$

After multiplying by the exact identity weight,

$$\int_{\rho_*}^1 \frac{D_H(t(\rho))}{n^2 \omega_n^{2/n} m(t(\rho))^{1-2/n}} (-t_\rho(\rho)) d\rho \geq c_n \int_{\rho_*}^1 \rho^{n+1} \|\partial_\rho h(\rho)\|_2^2 d\rho - \text{errors}.$$

This is the term missing from the one-level replacement argument.

5 The Sharp Conditional Theorem

Theorem. Fix $\rho_* \in (0, 1)$. There are constants $\varepsilon_0, c, C > 0$, depending only on n and ρ_* , with the following property. Suppose the outer level sets of Ω admit the coherent foliation gauge above on $[\rho_*, 1]$, with graph size at most ε_0 , and suppose the quadratic lower bounds in Sections 3 and 4 hold with absorbable errors. Then

$$\mathcal{A}(\Omega)^2 \leq C \delta_T(\Omega).$$

Proof. Add the lower bounds from Sections 3 and 4 and use the exact level-set identity restricted to the interval $\rho \in [\rho_*, 1]$. This gives (F). The trace estimate (T) gives

$$\|h(1)\|_2^2 \leq C \delta_T(\Omega).$$

Finally, the standard near-spherical comparison between the Fraenkel asymmetry and the L^2 boundary graph, after volume and translation normalization, gives

$$\mathcal{A}(\Omega)^2 \leq C \|h(1)\|_2^2.$$

6 What Must Replace the Selection Principle

The selection principle should be replaced by the following level-set entry statement, not by a stronger one-level Serrin theorem.

Outer foliation entry lemma. If $\delta_T(\Omega)$ is sufficiently small, then the actual torsion levels E_ρ , $\rho \in [\rho_*, 1]$, admit a modulated weak graph representation

$$\partial E_\rho = z(\rho) + (\rho + h(\rho, \theta))\theta$$

with

$$h \in H_\rho^1 L_\theta^2 \cap L_\rho^2 H_\theta^1$$

on the non-neutral modes, and the action estimate (F) holds.

This is a quantitative theorem about the original level sets. It does not construct a penalized minimizer. The proposed proof is:

1. Use the exact identity to control the total $D_I + D_H$ action on the fixed outer annulus $\rho \in [\rho_*, 1]$.
2. Use quantitative isoperimetry level by level to modulate each E_ρ by a center $z(\rho)$ and remove the zero and first modes.
3. Use the exact velocity formula for D_H to control the L_ρ^2 derivative of the non-neutral part of this modulation.
4. Apply the Hilbert-space trace theorem to recover the boundary graph.

This is the precise sense in which the global level-set data can act as the surrogate for the selection principle. The one-level route loses $1/\eta$; the foliation route replaces that average by radial kinetic energy.

7 Immediate Next Proof Obligation

The next concrete task is to prove the two quadratic lower bounds with all remainders:

$$D_I(t(\rho)) \gtrsim \rho^{2n-4} Q(h(\rho)),$$

and

$$D_H(t(\rho)) \gtrsim \rho^{2n-2} \|\partial_\rho h_{\geq 2}(\rho)\|_2^2 - C \rho^{2n-4} Q(h(\rho)) \|h\|_{C^1}.$$

These estimates are local computations in the moving graph gauge. Once written carefully, the only genuinely global issue is the outer foliation entry lemma.